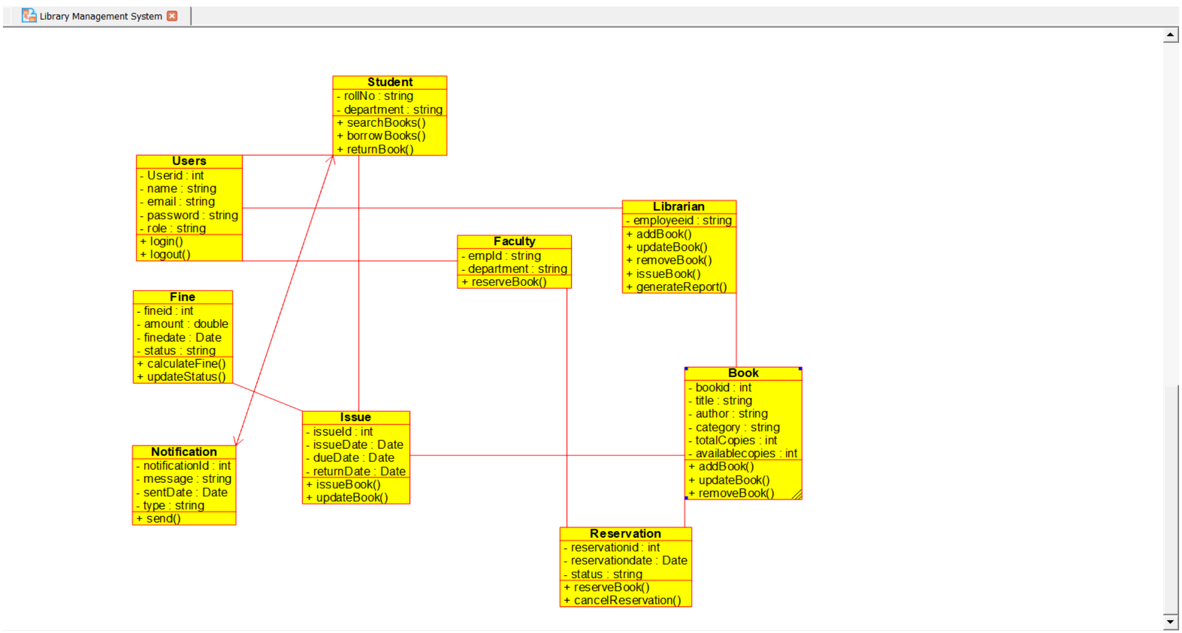
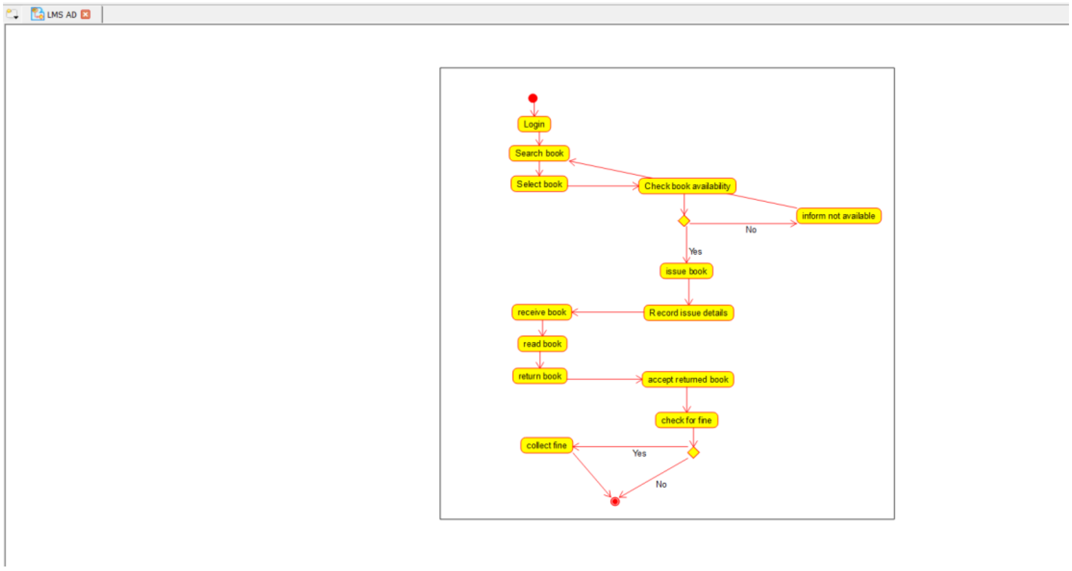
Question 1: Library Management System – Scenario: A college library currently maintains book issue and return records manually. Students often face delays while borrowing books because the librarian has to verify availability using registers. Faculty members can reserve books in advance, while librarians can add, update, and remove books from the catalog. Students can search books by title, author, or category. The librarian also calculates overdue fines and generates monthly reports. The college administration wants an automated system that provides faster services and minimizes manual errors. The system should also allow users to log in securely. Notifications should be sent for overdue books. Different users should have different access rights. As a system analyst, identify the actors and their interactions.

USE CASE DIAGRAM:

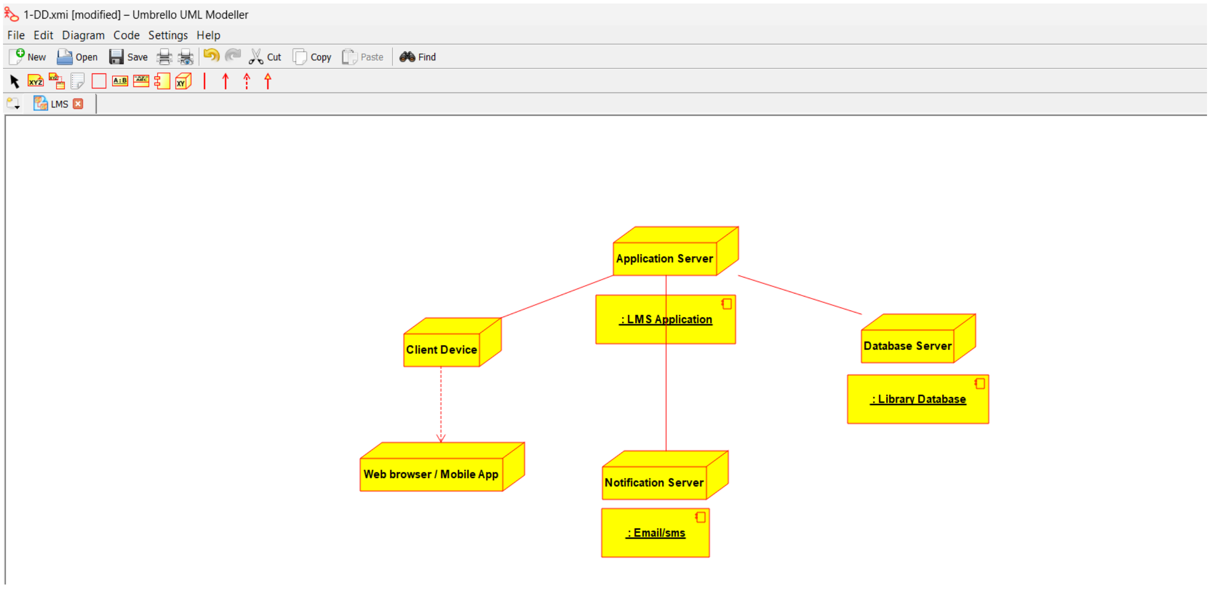
CLASS DIAGRAM:



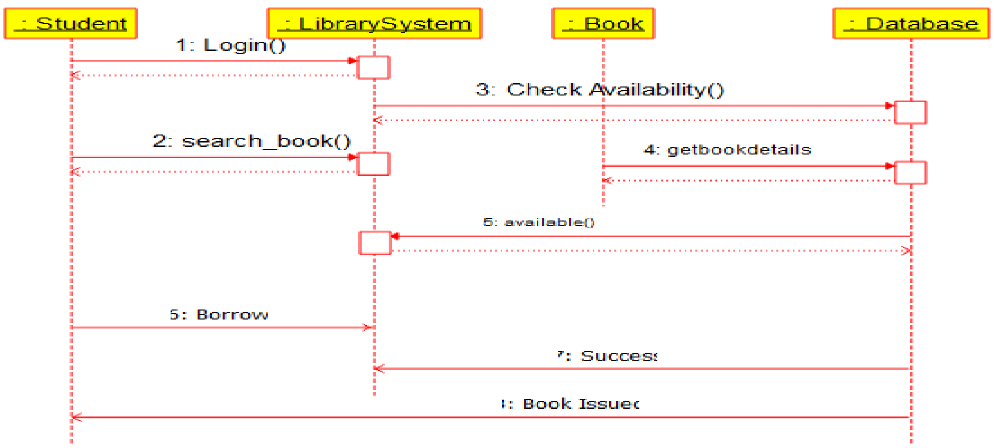
ACTIVITY DIAGRAM:



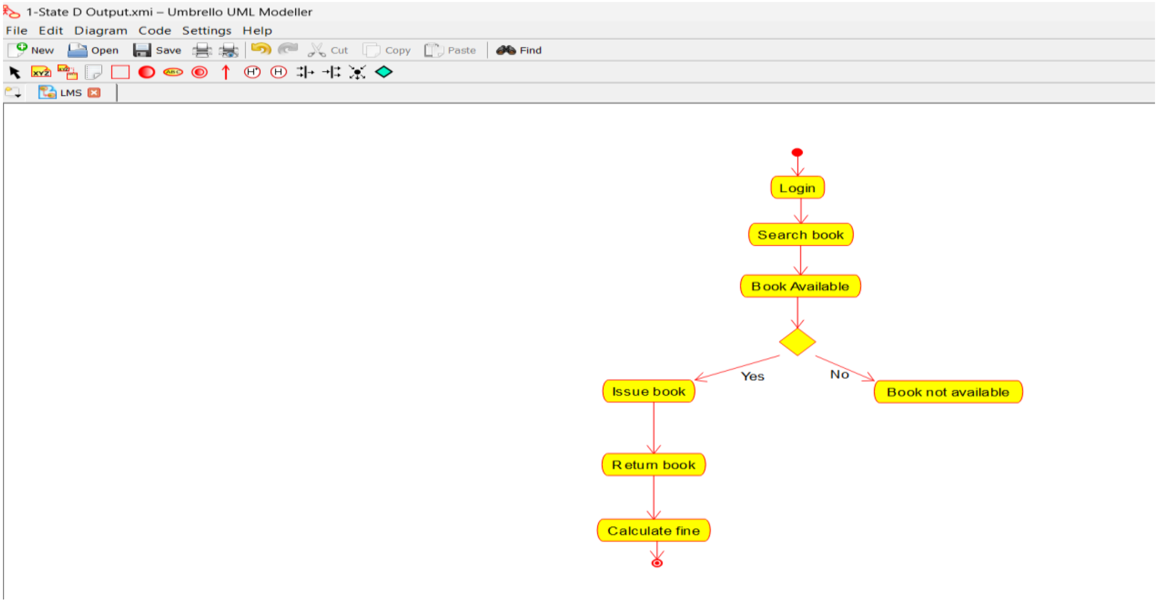
DEPLOYMENT DIAGRAM:



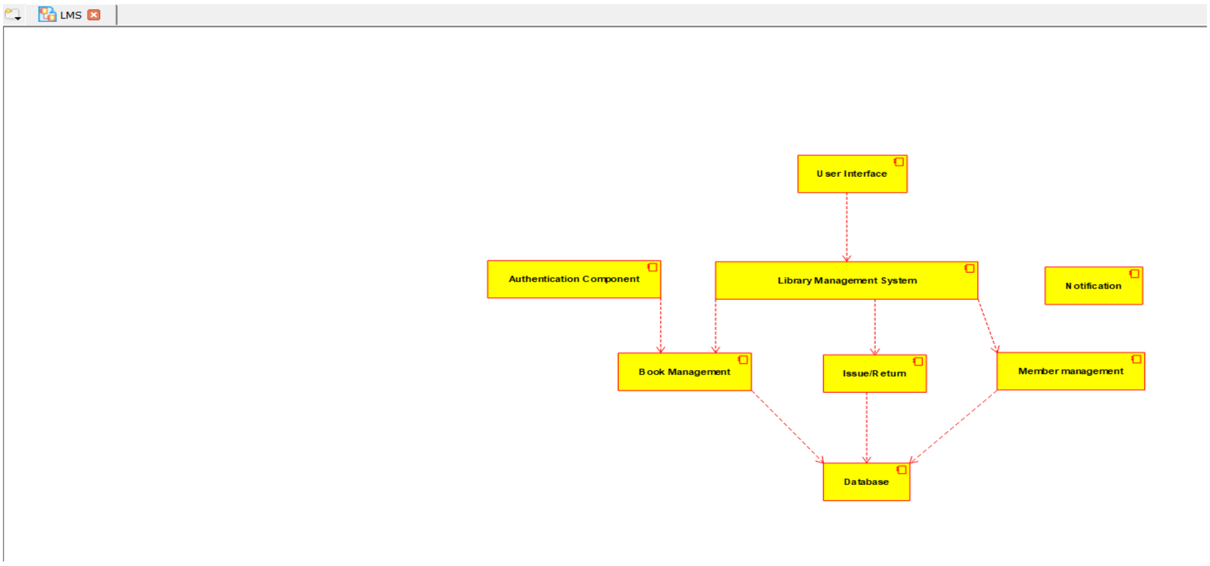
SEQUENCE DIAGRAM:



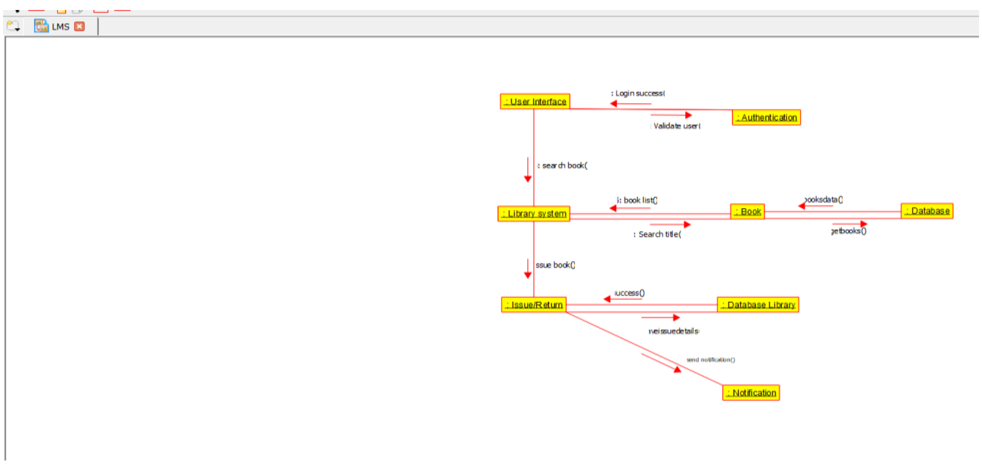
STATE MACHINE DIAGRAM:



COMPONENT DIAGRAM:



COMMUNICATION DIAGRAM:



Question 2: Hospital Management System – Class Diagram Scenario: A multispecialty hospital wants to automate patient management. Patients can register online, schedule appointments, and access medical reports. Doctors diagnose patients and prescribe medicines. Receptionists manage appointments and billing. Pharmacists issue medicines based on prescriptions. The billing department calculates treatment charges and insurance claims. Each patient has a unique medical history. Doctors belong to different departments. The hospital administrator manages staff information. Design the object-oriented structure for this system.

USE CASE DIAGRAM: